| | | | | |
|---|---|---|---|---|
| **2** | **(a)** | **(i)** $F = R\cos\theta$ | M1 | |
| | | $W = R\sin\theta$ | M1 | |
| | | dividing, $W = F\tan\theta$ | A0 | [2] |
| | | *(max. 1 if derivation to final line not shown)* | | |
| | | **(ii)** <u>provides</u> the centripetal force | B1 | [1] |
| | **(b)** | *either* $F = mv^2/r$ and $W = mg$ | | |
| | | *or* $v^2 = rg/\tan\theta$ | C1 | |
| | | $v^2 = (14 \times 10^{-2} \times 9.8)/\tan 28°$ | C1 | |
| | | $= 2.58$ | | |
| | | $v = 1.6\,\mathrm{m\,s^{-1}}$ | A1 | [3] |